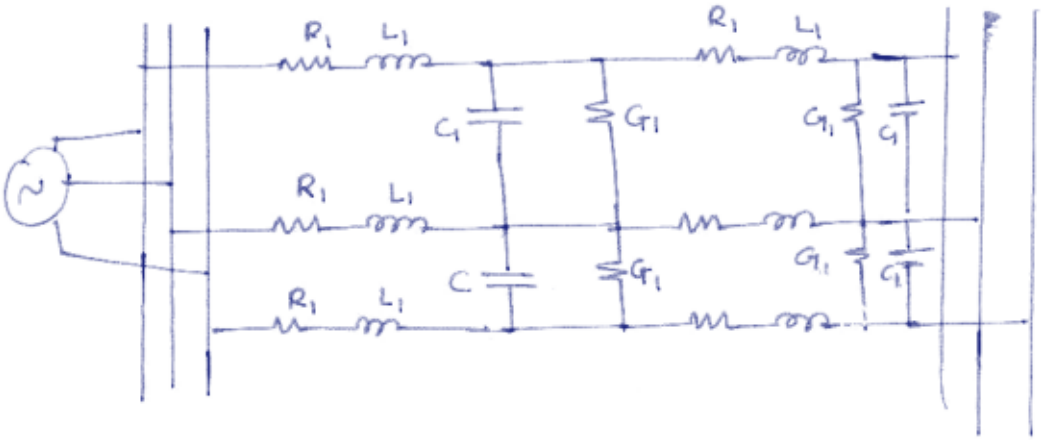
Transmission Line Performance

(*) This value is very small and hence neglected in case of short and medium line and considered in long transmission line.

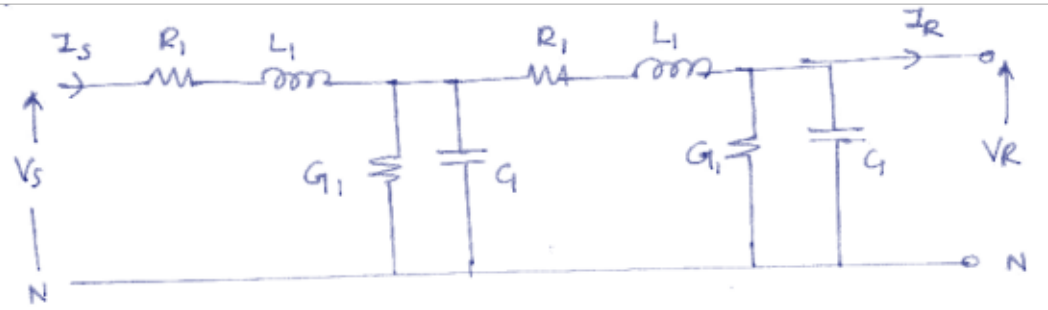
Modelling of T/Line



Three-phase line: R1, L1 in each phase, shunt C1 and G1 between phases and to neutral

Assumptions : (1) system is balanced: $I_R + I_Y + I_B = 0$, $V_R + V_Y + V_B = 0$

(R, Y, B)



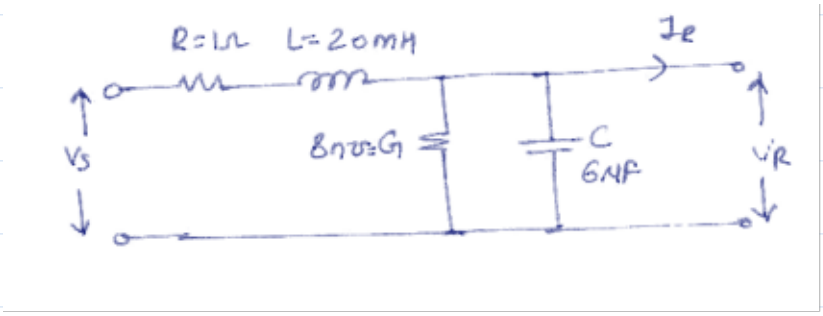
Distributed N/W [Long Tr. line equivalent ckt]: R1, L1 series; G1, C1 shunt; Vs, Is sending end, VR, IR receiving end

(2) Lumped N/W :

Assume l = 100 km

$R_1 = 0.01\,\Omega/km/ph$, $L_1 = 0.2\ mH/km/ph$, $C_1 = 0.06\ \mu F/km/ph$, $G_1 = 0.08\ n\mho/km/ph$

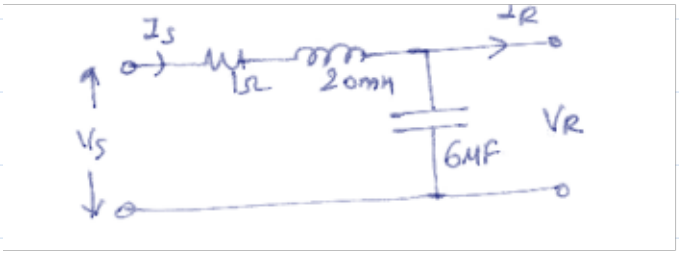
$R = R_1 \times l = 0.01 \times 100 = 1\,\Omega$, $L = L_1 \times l = 20\ mH$, $C = C_1 l = 6\ \mu F$, $G = G_1 l = 8\ n\mho$



Lumped model: R = 1 ohm, L = 20 mH in series; G = 8 n-mho and C = 6 uF in shunt

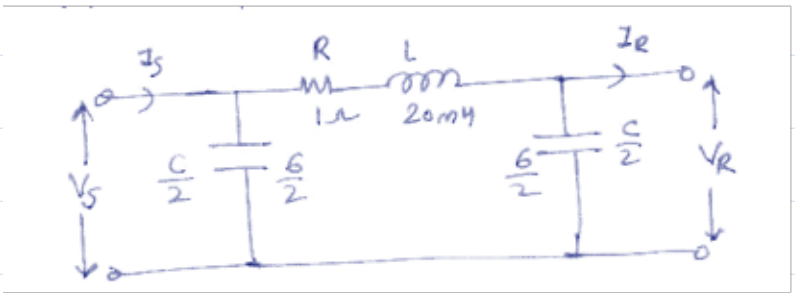
(3) Medium line

(a) End condenser N/W : assume G is neglected



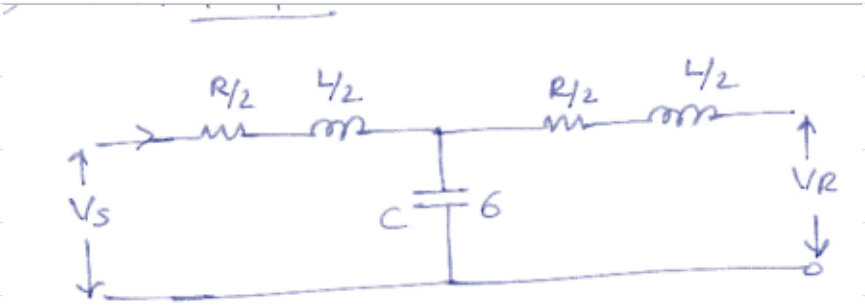
R = 1 ohm, L = 20 mH series; C = 6 uF shunt at receiving end

(b) Nominal π N/W :



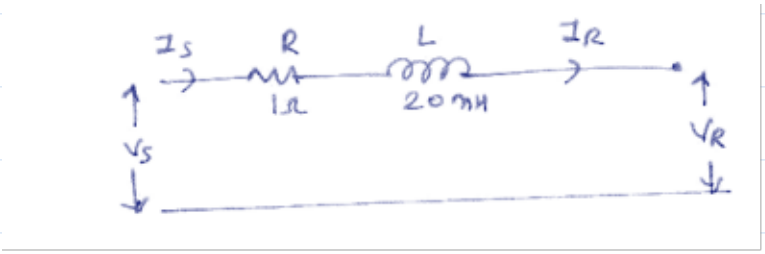
C/2 = 6/2 uF at both ends; R = 1 ohm, L = 20 mH series

(c) Nominal T - N/W :



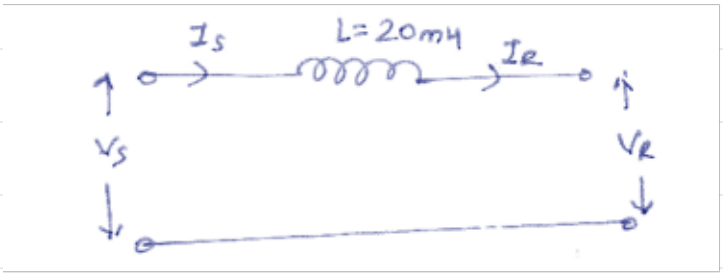
R/2, L/2 on either side of the shunt C = 6 uF

(4) Short transmission line : (*) (Neglect C and G)



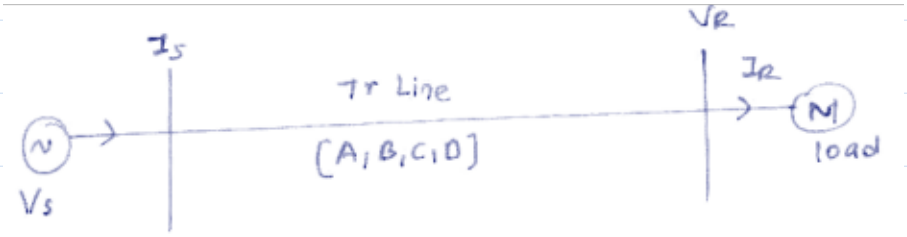
R = 1 ohm, L = 20 mH in series only

* Fault Analysis ($R \lll X_L$)



Only L = 20 mH in series

ABCD Tr. line parameters :



Source Vs, Is -> Tr line [A, B, C, D] -> VR, IR -> load

$$\begin{bmatrix} V_s \\ I_s \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix}$$

A = D => Symmetrical

AD - BC = 1 => Reciprocal.

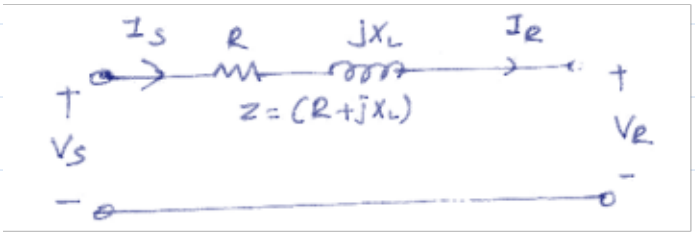
Cascading N/W :



Two networks [A1 B1; C1 D1] and [A2 B2; C2 D2] in cascade

$$\begin{bmatrix} V_s \\ I_s \end{bmatrix} = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix}$$

(1) Short Tr. Line :



Z = R + jXL in series; Vs, Is sending end, VR, IR receiving end

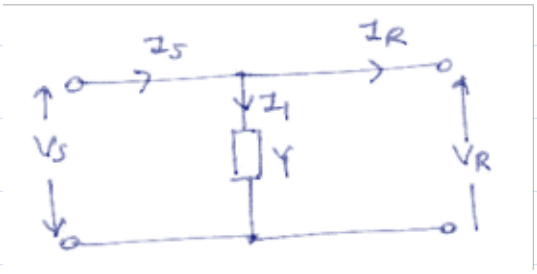
$$V_s = V_R + Z \cdot I_R = 1 \times V_R + Z \cdot I_R$$

$$I_s = I_R = 0 \times V_R + 1 \times I_R$$

$$\begin{bmatrix} V_s \\ I_s \end{bmatrix} = \begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix}$$

$$\boxed{Z = (R_1 + j\omega L_1)\,l}$$

(2) Admittance N/W :



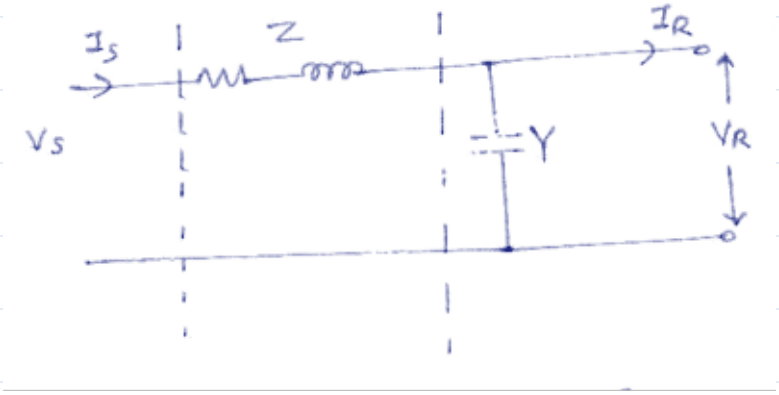
Shunt admittance Y across the line; Is, IR, I1; Vs = VR

$$V_s = V_R = 1 \times V_R + 0 \times I_R$$

$$I_s = I_1 + I_R = YV_R + 1 \times I_R$$

$$\begin{bmatrix} V_s \\ I_s \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ Y & 1 \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix}$$

(3) End condenser N/W :



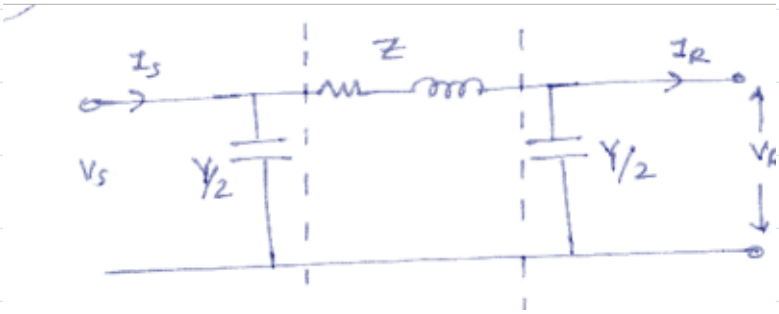
Z in series, then shunt Y at the receiving end

$$\begin{bmatrix} V_s \\ I_s \end{bmatrix} = \begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ Y & 1 \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix}$$

$$= \begin{bmatrix} (1+YZ) & Z \\ Y & 1 \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix}$$

$$Z = (R_1 + j\omega L_1)l, \qquad Y = j\omega C_1 l$$

(4) Nominal π N/W :

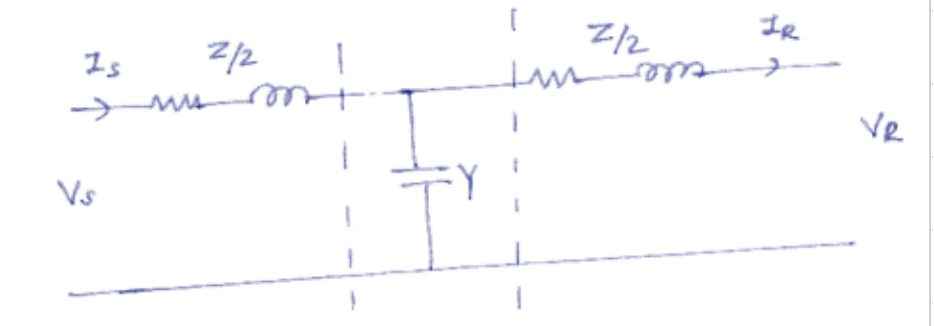


Y/2 at both ends, Z in series

$$\begin{bmatrix} V_s \\ I_s \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ Y/2 & 1 \end{bmatrix} \begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ Y/2 & 1 \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix}$$

$$= \begin{bmatrix} \left(1+\frac{YZ}{2}\right) & Z \\ Y\left(1+\frac{YZ}{4}\right) & \left(1+\frac{YZ}{2}\right) \end{bmatrix}$$

(5) Nominal T N/W :



Z/2 on each side of the shunt Y

$$\begin{bmatrix} V_s \\ I_s \end{bmatrix} = \begin{bmatrix} 1 & Z/2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ Y & 1 \end{bmatrix} \begin{bmatrix} 1 & Z/2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix}$$

$$= \begin{bmatrix} \left(1+\frac{YZ}{2}\right) & Z\left(1+\frac{YZ}{4}\right) \\ Y & \left(1+\frac{YZ}{2}\right) \end{bmatrix}$$

(6) Long T/L

$$\begin{bmatrix} V_s \\ I_s \end{bmatrix} = \begin{bmatrix} \cosh(\gamma l) & Z_c \sinh(\gamma l) \\ \frac{\sinh(\gamma l)}{Z_c} & \cosh(\gamma l) \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix}$$

$$\gamma = \alpha + j\beta = \sqrt{ZY}$$

α = Attenuation const.

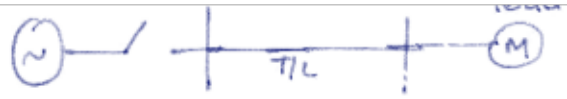
β = phase shift const.

γ = propagation const.

$$Z_c = \sqrt{\frac{Z}{Y}} = \sqrt{\frac{R_1 + j\omega L_1}{G_1 + j\omega C_1}} = \text{Ch. Impedance}$$

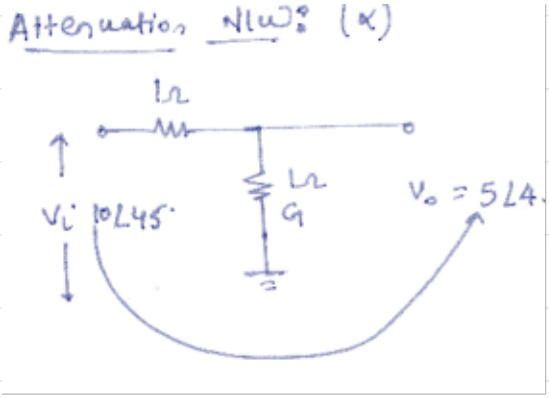
DAY-4 [11/9/25]

Meaning of α, β, γ



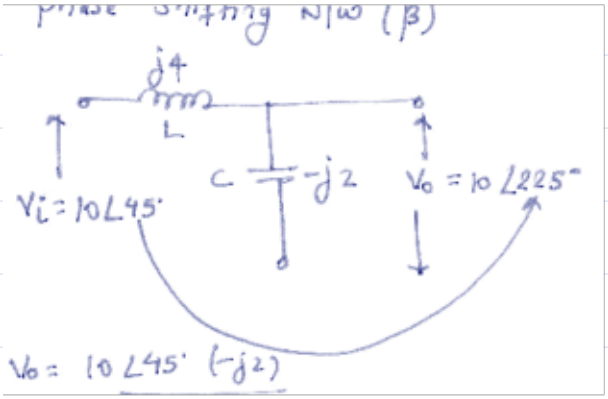
Generator - switch - T/L - load

Attenuation N/W : (α)



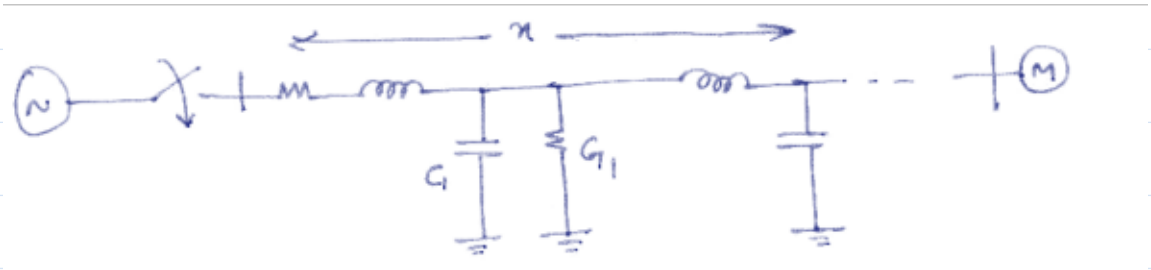
Series 1 ohm, shunt 1 ohm: Vi = 10 at 45 degrees, Vo = 5 at 45 degrees

Phase shifting N/W (β)



Series j4 (L), shunt -j2 (C): Vi = 10 at 45 deg, Vo = 10 at 225 deg

$$V_o = \frac{10\angle 45^\circ (-j2)}{j4 - j2} = 10\angle 225^\circ$$



Gen - switch - line with series R, L and shunt C1, G1 at distance x - load

$$V = V_m \sin(\omega t \pm \gamma x) = \text{Travelling wave eq}^n, \quad \gamma = (\alpha + j\beta)$$



Decaying sinusoidal wave

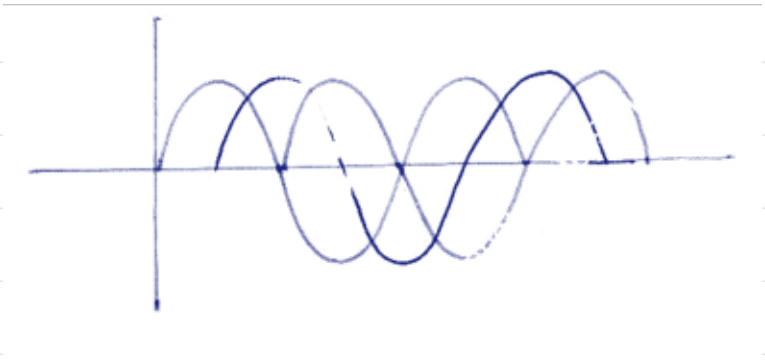
$V = V_m \sin(\omega t - \gamma x)$

γx V

(1) $0°$ $V_m \sin(\omega t)$

(2) $90°$ $-V_m \cos(\omega t)$

(3) $180°$ $-V_m \sin(\omega t)$



Wave at gamma x = 0, 90, 180 degrees

Velocity of wave : (v)

$$L_1 C_1 = \frac{H}{m} \cdot \frac{F}{m} \times \frac{\Omega}{\Omega} = \frac{H}{\Omega} \times F\Omega \times \frac{1}{m^2} = \frac{s^2}{m^2} = \frac{1}{v^2}$$

$$L_1 = \frac{\mu_0 \mu_r}{2\pi} \ln\left(\frac{d}{r}\right) \frac{H}{m}, \qquad C_1 = \frac{2\pi \varepsilon_0 \varepsilon_r}{\ln\left(\frac{d}{r}\right)} \frac{F}{m}$$

Assume $r' = r$

$$v = \frac{1}{\sqrt{L_1 C_1}} = \frac{1}{\sqrt{\mu_0 \mu_r \varepsilon_0 \varepsilon_r}} = \frac{3 \times 10^8}{\sqrt{\varepsilon_r \mu_r}} = \frac{2\pi f L}{\sqrt{B \cdot X_L}}$$

X_L = Total Inductive Reactance of line = $2\pi f L_1 l$

B = Total capacitive Susceptance of line = $2\pi f C_1 l$

Ideal T/L : $v = 3 \times 10^8$ m/sec = 300 m/μs

practical overhead T/L => 200 - 250 m/μs

Cable ($\varepsilon_r \gg 1$) => 40 - 50 m/μsec

Lossless / Loss free / Ideal Tr. line

A T/L in which $R_1 = G_1 = 0$ indicates no power loss.

α, β for lossless T/L :

$$\gamma = \alpha + j\beta = \sqrt{Z_1 Y_1} = \sqrt{(R_1 + j\omega L_1)(G_1 + j\omega C_1)}$$

Substitute $R_1 = G_1 = 0$

$$\alpha + j\beta = j\omega\sqrt{L_1 C_1} \Rightarrow \alpha = 0$$

$$\beta = \omega\sqrt{L_1 C_1}, \quad \beta = \frac{\omega}{v} = \frac{2\pi f}{v} \text{ rad/unit length} = \frac{360^\circ f}{v} \text{ Degree/unit length}$$

Electrical wavelength (β')

l = length of T/L

$$\beta' = \beta l \text{ rad} = \frac{2\pi f l}{v}$$

$v = f \cdot \lambda$, λ = wavelength of signal

$$\beta' = \frac{2\pi f l}{f\lambda} \Rightarrow \frac{l}{\lambda} = \frac{\beta'}{2\pi}$$

$$\frac{\text{physical length of T/L}}{\text{wavelength of signal}} = \frac{\beta'}{2\pi}$$

Ideal Parameters : $f = 50\ Hz$, $v = 3 \times 10^8\ m/s$

$$\beta' = \frac{2\pi \times 50}{3 \times 10^8} l \text{ rad}$$

A = D of lossless T/L

From Nominal π N/W :

$$A = D = \left(1 + \frac{YZ}{2}\right) = 1 + \frac{(j\omega C_1 l)(j\omega L_1 l)}{2} = 1 - \frac{\omega^2 L_1 C_1 l^2}{2}$$

$$A = D = 1 - \frac{(\beta')^2}{2}$$

where β' is in rad

Surge / Lossless / flat / Infinite / Resonant / Tuned / Natural line / Ideal line Impedance (Z_s)

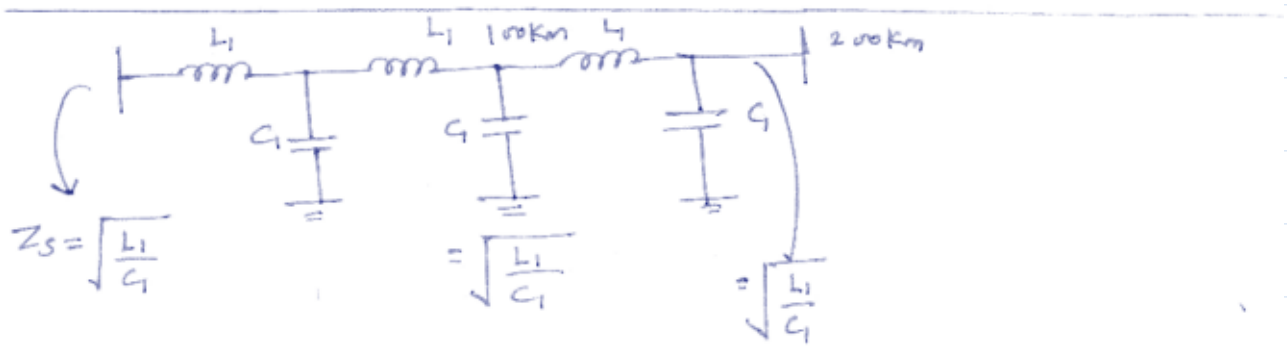
$$Z_c = \sqrt{\frac{Z_1}{Y_1}} = \sqrt{\frac{R_1 + j\omega L_1}{G_1 + j\omega C_1}}$$

Substitute $R_1 = G_1 = 0$ for lossless T/L

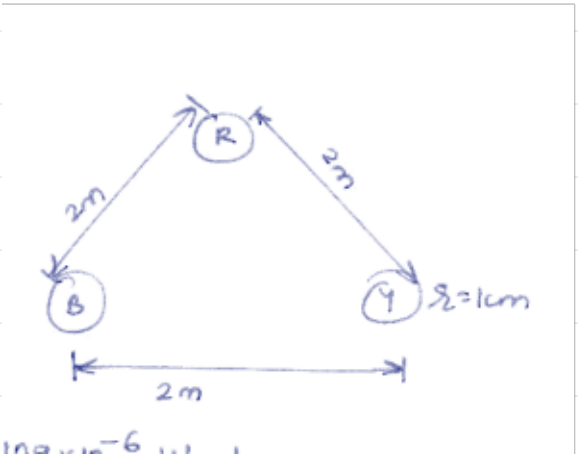
$$Z_c = \sqrt{\frac{L_1}{C_1}} = Z_s$$

Z_s = 300 - 400 Ω for overhead T/L ; = 40 - 50 Ω for cable

NOTE : Z_s is independent of length of T/L



Lossless line of 200 km: L1, C1 sections; the impedance Zs = sqrt(L1/C1) looking in at any point



Equilateral arrangement R, Y, B of 2 m sides, r = 1 cm

$$L_1 = 1.109 \times 10^{-6}\ H/m/ph, \quad C_1 = 10.49 \times 10^{-12}\ F/m/ph$$

$$Z_s = \sqrt{\frac{1.109 \times 10^{-6}}{10.49 \times 10^{-12}}} \approx 325\,\Omega$$

$$v = \frac{1}{\sqrt{1.109 \times 10^{-6} \times 10.49 \times 10^{-12}}} = 2.93 \times 10^{8}\ m/sec$$

* A Tr. line which is independent of frequency, i.e. which allows all the frequency, is called distortionless line.

* It is practically used in communication application (Dish cable).

* If the series element ($R + jX_L$) time const. i.e. L/R is equal to the shunt element (C, G) time const. [T.C = C/G], then line becomes independent of freq.

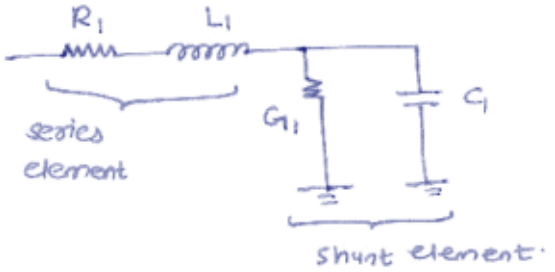
$$Z_c = \sqrt{\frac{R_1 + j\omega L_1}{G_1 + j\omega C_1}} \leftarrow \text{depends on frequency}$$

$$= \sqrt{\frac{R_1}{G_1}} \sqrt{\frac{\left(1 + j\frac{\omega L_1}{R_1}\right)}{\left(1 + j\frac{\omega C_1}{G_1}\right)}}$$

If $\frac{L_1}{R_1} = \frac{C_1}{G_1}$ (condition for distortionless T/L)

$$\boxed{Z_c = \sqrt{\frac{R_1}{G_1}} = Z_D} \leftarrow \text{Independent of frequency}$$

Z_D = Distortionless line Impedance



Series element R1, L1; shunt element G1, C1

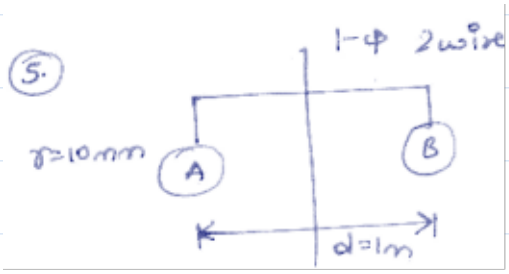
4] $L_{circuit} = \frac{\mu_0}{\pi} \ln \frac{d}{r'}$ H/m

$$= 4 \times 10^{-7} \ln \frac{1.5}{0.7788 \times 0.5 \times 10^{-2}} = 2.38 \times 10^{-6} \text{ H/m}$$

$l = 10$ km

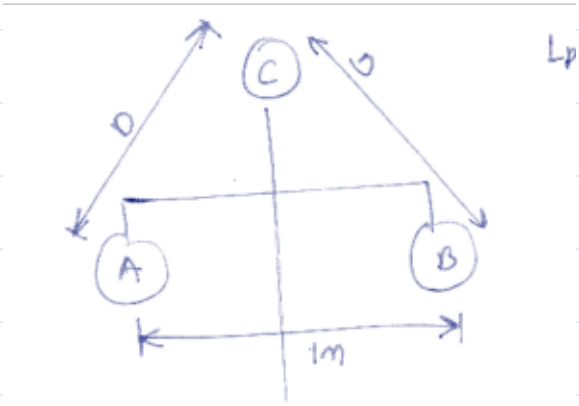
$L = 2.38 \times 10^{-6} \times 10 \times 10^3 = 23.8$ mH

(5) 1-ϕ 2-wire $r = 10$ mm, $d = 1$ m



Conductors A, B of radius 10 mm, spacing d = 1 m

$$L_A = L_B = 2\times10^{-7}\ln\left(\frac{1}{0.7788\times10\times10^{-3}}\right) = 0.97\times10^{-6}\ H/m$$



Third conductor C added above, A and B 1 m apart, distance D from C

$$L_{ph} = 2\times10^{-7}\ln\frac{(D\cdot D\times1)^{1/3}}{0.7788\times10\times10^{-3}} = 1.05\times L_A = 1.05\times0.97\times10^{-6}\quad (5\%\ \text{increase})$$

$$D = 1.439$$

Vtg Regulation :

$$\%V_{REG} = \frac{V_{RNL}-V_{RFL}}{V_{RFL}}\times100 = \frac{\left|\frac{V_S}{A}\right| - |V_R|}{|V_R|}\times100$$

. V_{RNL} = No load receiving end vtg. At no load $I_R = 0$

$$V_S = AV_R + B(0) = AV_{RNL} \rightarrow V_{RNL} = \frac{V_S}{A}$$

. V_{RFL} = Full load or (rated receiving end vtg) = V_R

-> Ideal Transmission line has vtg Regulation 0%.

-> Practically line ±5% Vtg Regulation in 220 kV line and up to ±10%.

Procedure for calculation of performance parameter of Tr. line.

(1) From the $P_{R3\phi}$, V_{RL}, $\cos\phi_R$ of T/L (assume star load) :

. Calculate

$$|I_R| = \frac{P_{R3\phi}}{\sqrt{3}\, V_R \cos\phi_R}$$

(2) $I_R = |I_R| \angle \phi_R$; ϕ_R = +ve for lead load ; = -ve for lag load ; = 0 for UPF load.

(3) Calculate

$$V_{Sph} = A\frac{V_{RL}}{\sqrt{3}} + BI_R = |V_{Sph}| \angle \theta_1$$

(4)

$$V_{SL} = \sqrt{3}\, V_{Sph} = \sqrt{3}\, |V_{Sph}| \angle \theta_1$$

(5)

$$I_{SL} = I_{Sph} = C\frac{V_{RL}}{\sqrt{3}} + DI_R = |I_{SL}| \angle \theta_2$$

(6)

$$\phi_S = \theta_1 - \theta_2$$

(7)

$$P_{S3\phi} = \sqrt{3}\, V_{SL} I_{SL} \cos\phi_S$$

(8) Transmission Efficiency

$$\%\eta = \frac{P_{R3\phi}}{P_{S3\phi}} \times 100$$

(9) Transmission line loss

$$P_{L3\phi} = (P_{S3\phi} - P_{R3\phi})$$

(10) Vtg Regulation

$$\%V_{reg} = \frac{\left|\frac{V_S}{A}\right| - |V_R|}{|V_R|} \times 100$$

Eg: Per phase parameters of 50 Hz, 100 km long, 3-ϕ Tr. line are $10\,\Omega$, $L = 111.7$ mH, $C = 0.9954\,\mu F$. The Transmission line delivers 20 MW at a p.f of 0.8 lag and 66 kV to a balanced load. Calculate the sending end vtg, η, and regulation using nominal T-representation.

$$Z = 10 + j(2\pi \times 50 \times 111.7 \times 10^{-3}) = 36.49\angle 74^\circ$$

$$Y = j(2\pi \times 50 \times 0.9954 \times 10^{-6}) = 0.313 \times 10^{-3}\angle 90^\circ$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \left(1 + \frac{YZ}{2}\right) & Z\left(1 + \frac{YZ}{4}\right) \\ Y & \left(1 + \frac{YZ}{2}\right) \end{bmatrix}$$

$$= \begin{bmatrix} 0.994\angle 0.09^\circ & 36.389\angle 74.14^\circ \\ 0.313 \times 10^{-3}\angle 90 & 0.994\angle 0.09^\circ \end{bmatrix}$$

$$|I_R| = \frac{P_{R3\phi}}{\sqrt{3}\,V_R \cos\phi_R} = \frac{20 \times 10^6}{\sqrt{3} \times 66 \times 10^3 \times 0.8} = 218.7\ A$$

$$I_R = 218.7\angle -36.86^\circ\ A$$

$$V_{Sph} = A\frac{V_{RL}}{\sqrt{3}} + BI_R = 0.994\angle 0.09^\circ \times \frac{66 \times 10^3}{\sqrt{3}} + 36.369\angle 74.14^\circ \times 218.7\angle -36.86^\circ$$

$$= 44.45\angle 6.38^\circ\ kV/ph$$

$$V_{SL} = \sqrt{3}\,V_{Sph} = 77.03\angle 6.38^\circ\ kV$$

$$I_{SL} = I_{Sph} = C\frac{V_{RL}}{\sqrt{3}} + DI_R = 0.313 \times 10^{-3}\angle 90^\circ \times \frac{66 \times 10^3}{\sqrt{3}} + 0.994\angle 0.09^\circ \times 218.7\angle -36.86^\circ$$

$$= 210.4\angle -34.17^\circ$$

$$\phi_S = 6.38^\circ - (-34.17) = 40.55$$

$$P_{S3\phi} = \sqrt{3}\, V_{SL} I_{SL} \cos\phi_S = \sqrt{3} \times 77.03 \times 210.4 \cos(40.55) = 21.325\ MW$$

$$\%\eta = \frac{P_{R3\phi}}{P_{S3\phi}} \times 100 = \frac{20}{21.325} \times 100 = 93.78\%$$

$$\%V_{reg} = \frac{\left|\frac{V_S}{A}\right| - |V_R|}{|V_R|} \times 100 = \frac{\left|\frac{77.03}{0.994}\right| - |66|}{|66|} \times 100 = 17.45\%$$

Vtg Regulation of line w/o compensation

(1) Short Tr. line

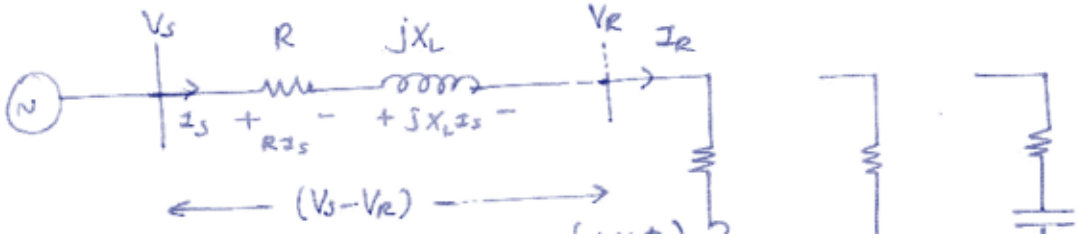
$\overline{|66|}$

$= 17.45\%$.

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3 Regulation of line w/o compensation

Short Tr. line



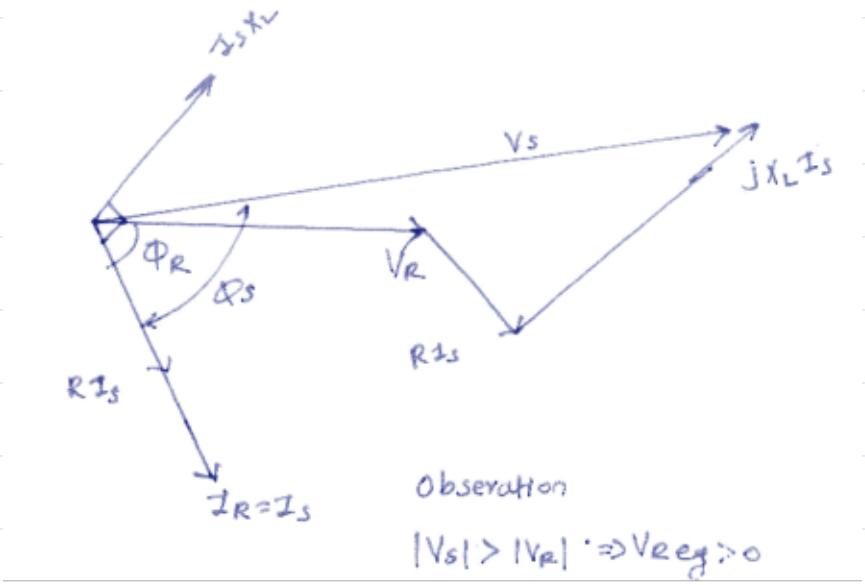
Generator Vs - R - jXL - VR, IR; lag load (fans, motors), UPF load, lead load (music chargers): Vs - VR = R Is + j Is XL

$$\overline{V_s} = \overline{V_R} + R\overline{I_s} + j\overline{I_s}\overline{X_L}$$

$$\%V_{Reg} = \frac{\left|\frac{V_S}{A}\right| - |V_R|}{|V_R|} \times 100\ ;\ \text{for short Tr. line } A = 1$$

$$\%V_{reg} = \frac{V_S - V_R}{V_R} \times 100$$

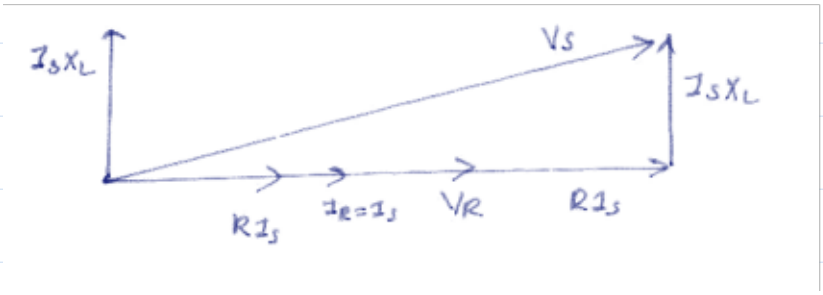
lag load :



Phasor diagram, lagging load: VR, R Is, j XL Is, Vs, angles phiR and phiS; IR = Is

Observation : $|V_S| > |V_R| \Rightarrow V_{Reg} > 0$

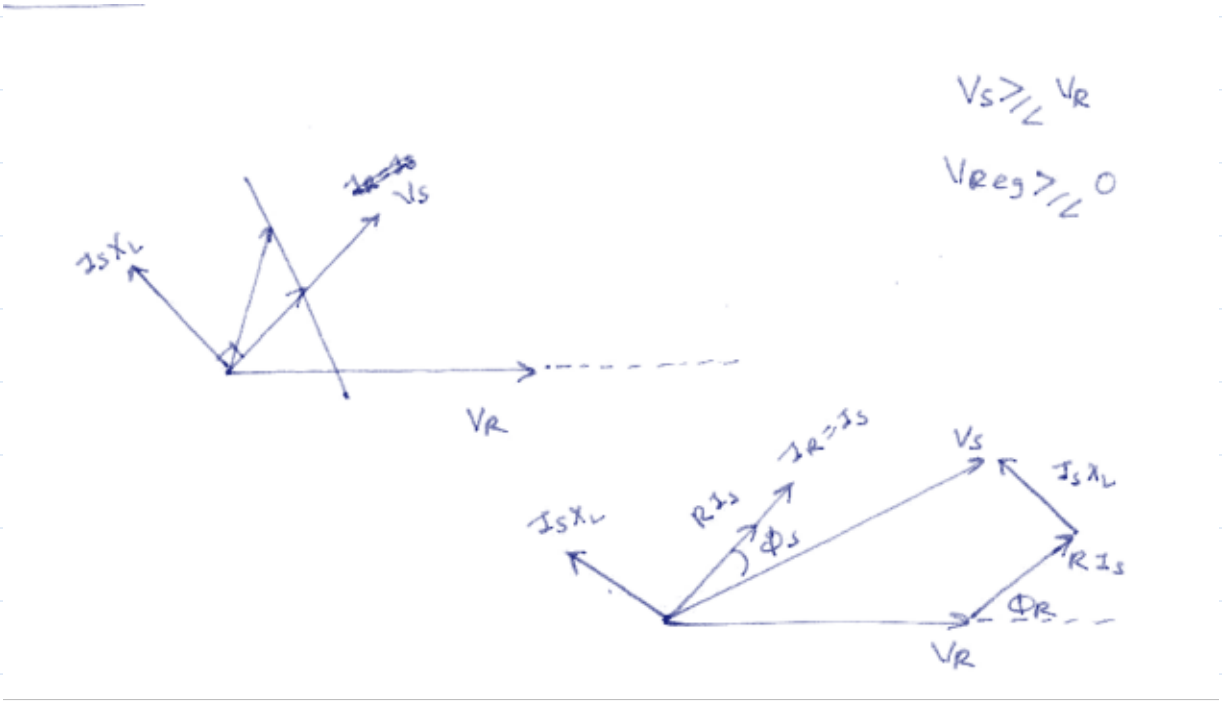
UPF load :



Phasor diagram, UPF load: IR = Is along VR, R Is, Is XL

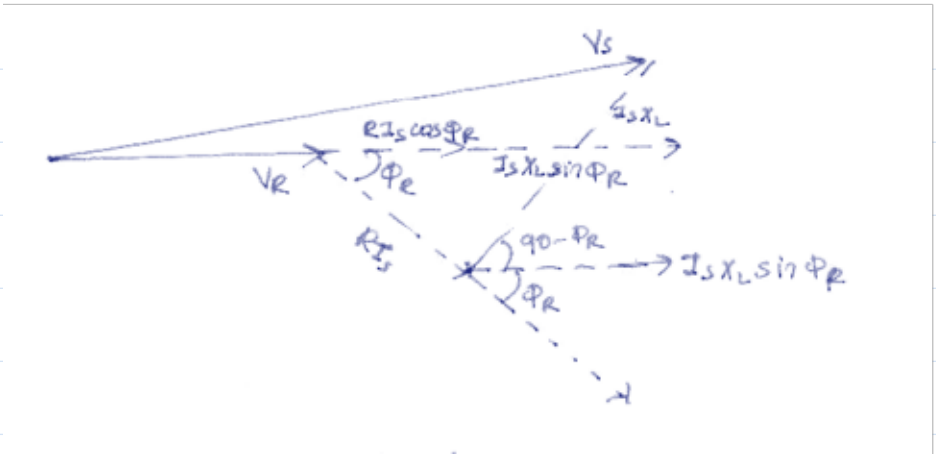
Observation : $V_S > V_R$; $V_{Reg} > 0$

lead load :



Phasor diagrams, leading load: Vs, VR, R Is, Is XL, phiR, phiS; Vs >/< VR; VReg >/< 0

Condition for Maximum Vtg Regulation with lagging load



Resolving R Is and Is XL along VR: R Is cos phiR and Is XL sin phiR

$$V_S = V_R + RI_s\cos\phi_R + I_sX_L\sin\phi_R$$

$$\boxed{V_S = V_R + I_s(R\cos\phi_R + X_L\sin\phi_R)}$$

$$\%V_{Reg} = \frac{V_S - V_R}{V_R} \times 100 = \frac{I_s(R\cos\phi_R + X_L\sin\phi_R)}{V_R} \times 100$$

$$\text{For Max } V_{reg}: \frac{d}{d\phi_R}(R\cos\phi + X_L\sin\phi_R) = 0$$

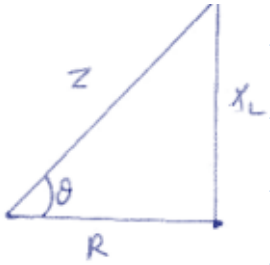
$$-R\sin\phi_R + X_L\cos\phi_R = 0 \Rightarrow \tan\phi_R = \frac{X_L}{R}$$

$$\boxed{\phi_R = \tan^{-1}\frac{X_L}{R} \Rightarrow \text{condition for Max } V_{reg}}$$

. Transmission line Impedance $Z = R + jX_L = |Z|\angle\theta$

. Transmission line Impedance Angle : $\theta = \tan^{-1}\frac{X_L}{R} = \phi_R$

$$\boxed{\text{power factor of lag load for Max Vtg Reg. : } \cos\phi_R = \cos\theta = \frac{R}{Z} \text{ lag.}}$$



Impedance triangle: R, XL, Z, angle theta

lead load. (ϕ_R = -ve)

$$V_S - V_R = I_s(R\cos\phi_R - X_L\sin\phi_R)$$

$$\%V_{Reg} = \frac{V_S - V_R}{V_R} \times 100 = \frac{I_s(R\cos\phi_R - X_L\sin\phi_R)}{V_R} \times 100$$

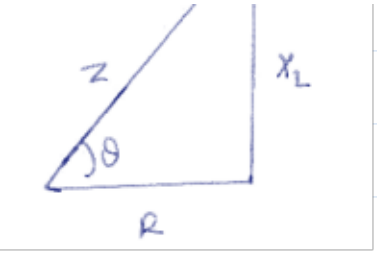
. For $V_{reg} = 0$: $(R\cos\phi_R - X_L\sin\phi_R) = 0 \Rightarrow \tan\phi_R = \frac{R}{X_L}$

$$\boxed{\phi_R = \tan^{-1}\frac{R}{X_L} \Rightarrow \text{condition for } V_{Reg} = 0}$$

Transmission line Impedance $Z = R + jX_L = |Z|\angle\theta$

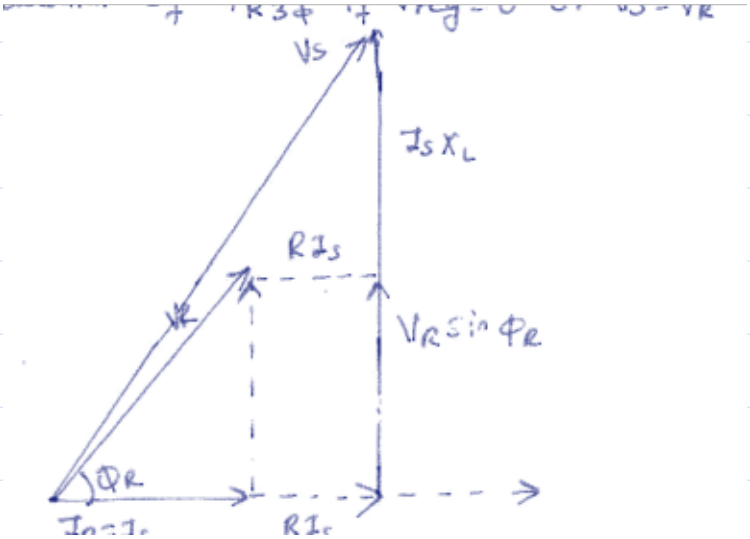
Transmission line Impedance Angle : $\phi_R = \frac{\pi}{2} - \theta = \tan^{-1}\frac{X_L}{R}$

power factor of lead load for $V_{reg} = 0$: $\cos\phi_R = \sin\theta = \frac{X_L}{Z}$ lead



Impedance triangle: R, XL, Z, angle theta

Calculation of $P_{R3\phi}$ if $V_{reg} = 0$ or $V_S = V_R$ is given



Phasor diagram: Vs = VR, Is XL, R Is, VR sin phiR, VR cos phiR, IR = Is, angle phiR

$$V_S^2 = (V_R\cos\phi_R + RI_s)^2 + (V_R\sin\phi_R + I_sX_L)^2$$

$$= V_R^2 + I_s^2(R^2 + X_L^2) + 2V_RI_s(R\cos\phi_R + X_L\sin\phi_R)$$

$$I_s^2(R^2 + X_L^2) \lll V_R^2$$

$$V_S = \left[V_R^2 + 2V_RI_s[R\cos\phi_R + X_L\sin\phi_R]\right]^{1/2} \text{ (using approximation)}$$

$$V_S = V_R + \ldots$$

$$I_s = I_R = \frac{\frac{2V_{RL}}{\sqrt{3}}(R\cos\phi_R + X_L\sin\phi_R)}{(R^2 + X_L^2)} \text{ - lag load}$$

$$I_s = I_R = \frac{\frac{2V_{RL}}{\sqrt{3}}(R\cos\phi_R - X_L\sin\phi_R)}{(R^2 + X_L^2)} \quad - \text{lead load}$$

$$V_R = V_{ph} = \frac{V_{RL}}{\sqrt{3}}$$

$$\boxed{P_{R3\phi} = \sqrt{3}\, V_{RL} I_R \cos\phi_R}$$

Eg: Find $P_{R3\phi}$ if $V_S = V_R = 11\ kV$; $Z = 3 + j4$; p.f = 0.8 lag.

$$I_s = I_R = \frac{\frac{2V_{RL}}{\sqrt{3}}(R\cos\phi_R + X_L\sin\phi_R)}{R^2 + X_L^2}$$

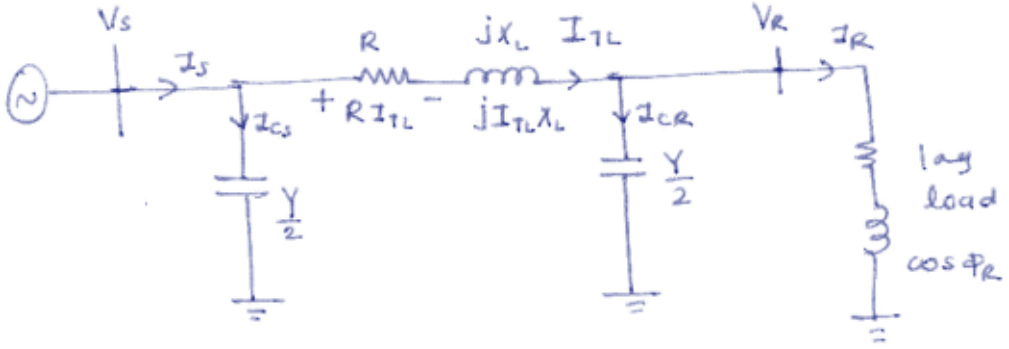
$$= \frac{2 \times \frac{11 \times 10^3}{\sqrt{3}}[3 \times 0.8 + 4 \times 0.6]}{3^2 + 4^2} = 2.438\ kA$$

$$P_{R3\phi} = \sqrt{3} \times 11 \times 2.438 \times 10^3 \times 0.8 = 37.71\ MW$$

DAY-5 [13/9/25]

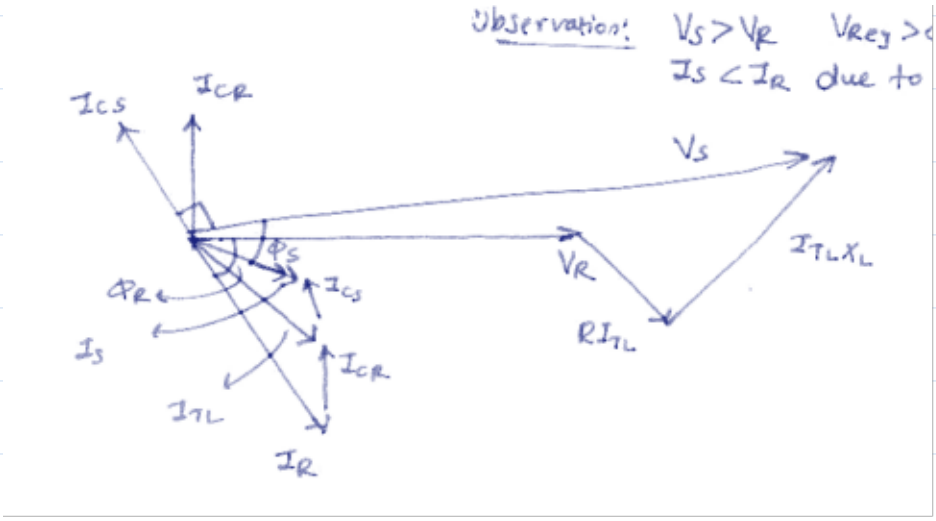
(a) Medium Tr. line

(a) Nominal π N/W :



Vs, Is - Y/2 (Ics) - R, jXL, ITL - Y/2 (ICR) - VR, IR - lag load; voltage drops R ITL, j ITL XL

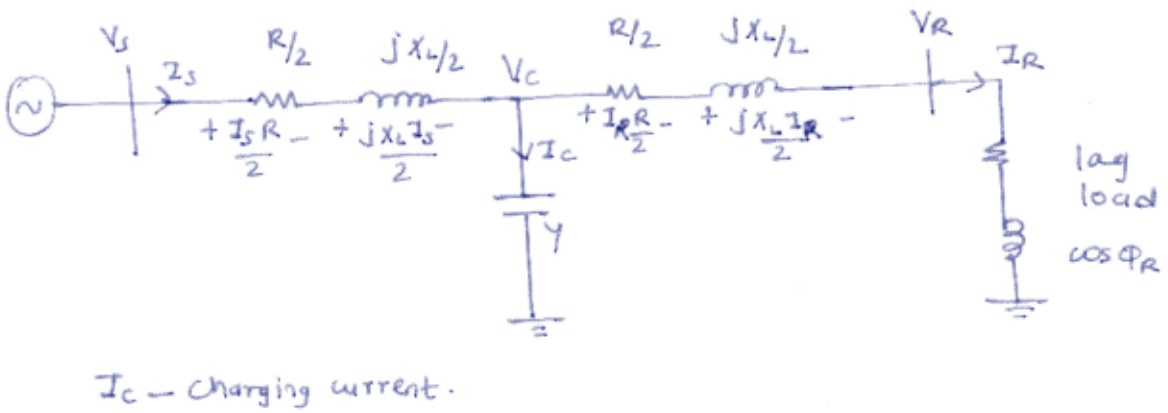
I_{CR}, I_{CS} - charging current ; I_{TL} -> current through TL



Phasor diagram: Vs, VR, R ITL, ITL XL, Ics, ICR, Is, ITL, IR, angles phiR and phiS

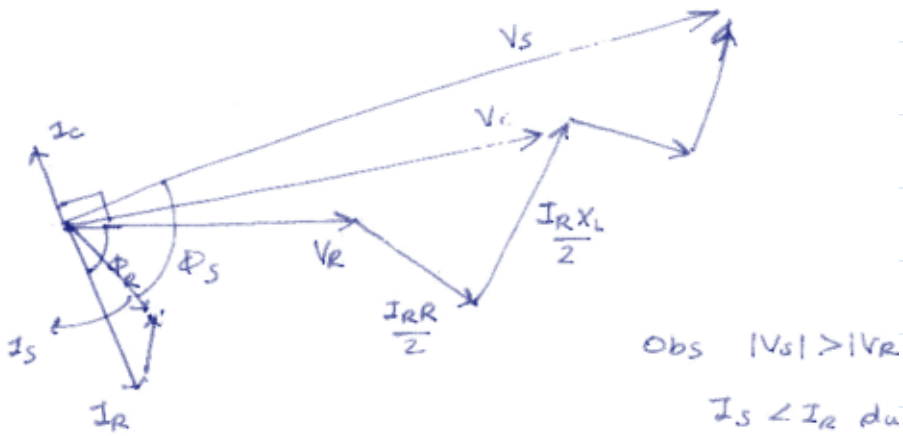
Observation : $V_S > V_R$, $V_{Reg} > 0$; $I_S < I_R$ due to I_{CR}, I_{CS}

(b) Nominal T N/W :



Vs, Is - R/2, jXL/2 - Vc, Ic, Y - R/2, jXL/2 - VR, IR - lag load

I_c - Charging current.



Phasor diagram: VR, IR R/2, IR XL/2, Vc, Ic, Is, Vs, angles phiR, phiS

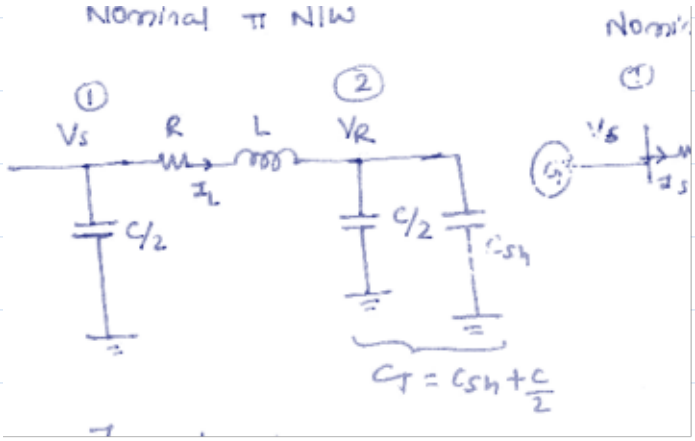
Obs : $|V_S| > |V_R|$, $V_{reg} > 0$; $I_S < I_R$ due to I_c

Comparison between short and Medium line

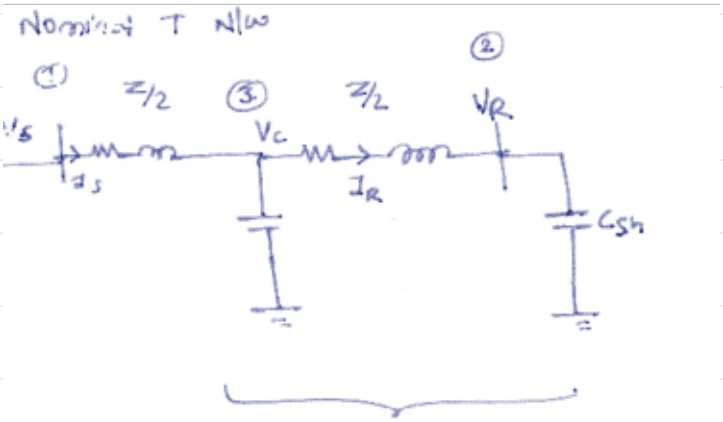
Assume both lines have same length

Parameter	Short	Medium
I_s	$I_s = I_R$	$I_s < I_R$ due to I_c
P_L	More	Less
Efficiency	Less	More
Vtg Reg	More	Less
P_s	More	Less

Comparison between π and T N/W :



Nominal pi N/W: nodes 1 (Vs) and 2 (VR); C/2 at both ends, Csh at receiving end; CT = Csh + C/2; IL = (Vs - VR)/Z



Nominal T N/W: nodes 1 (Vs), 3 (Vc), 2 (VR); Z/2 each side; Is = (Vs - Vc)/(Z/2), IR = (Vc - VR)/(Z/2)

Nominal π : (1) No. of nodes are only 2, so that no. of equations are less.

(2) It is easier to combine external compensator (shunt Capacitor, Inductor) with the line capacitance.

Nominal T : (1) No. of nodes are 3, so that no. of equations are more.

(2) It is complex, it is difficult.

(3) Due to above advantage π N/W is practically used as Tr. line equivalent ckt in Load flow studies.

(3) T N/W : practically not preferred.

(*) Do not apply Star Delta Transformation for conversion of T-N/W into π N/W and vice-versa.

Equivalent π N/W of long Line

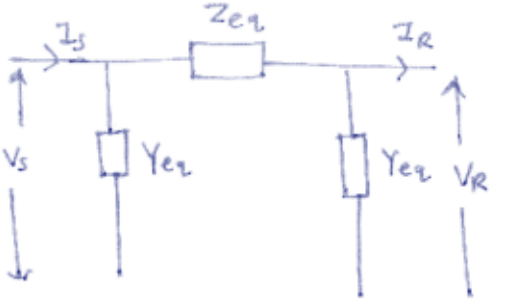
Long lines are normally represented as distributed N/W. For using this long line in the load flow studies, distributed N/W is converted into equivalent π N/W.

From Nominal π N/W

$$\begin{bmatrix} V_s \\ I_s \end{bmatrix} = \begin{bmatrix} \left(1+\frac{YZ}{2}\right) & Z \\ Y\left(1+\frac{YZ}{4}\right) & \left(1+\frac{YZ}{2}\right) \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix} \quad -(1)$$

From long line N/W :

$$\begin{bmatrix} V_s \\ I_s \end{bmatrix} = \begin{bmatrix} \cosh\gamma l & Z_c \sinh\gamma l \\ \frac{\sinh\gamma l}{Z_c} & \cosh\gamma l \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix} \quad -(2)$$



Equivalent pi: Zeq in series, Yeq at each end; Vs, Is, VR, IR

From (1) and (2) :

$$\boxed{Z = Z_{eq} = Z_c \sinh(\gamma l)}$$

$$\left(1+\frac{YZ}{2}\right) = \cosh(\gamma l)$$

$$\frac{Y}{2} = \frac{\cosh\gamma l - 1}{Z} = \frac{1+2\sinh^2\frac{\gamma l}{2}-1}{Z_c \sinh\gamma l}$$

$$= \frac{1+2\sinh^2\frac{\gamma l}{2}-1}{Z_c\left(2\sinh\frac{\gamma l}{2}\cosh\frac{\gamma l}{2}\right)}$$

$$\boxed{\frac{Y}{2} = \frac{\tanh\frac{\gamma l}{2}}{Z_c} = Y_{eq}}$$

Z_{eq} for loss less line {R = G = 0}

$$Z_{eq} = Z_c \sinh(\gamma l) = Z_c \sinh(\alpha l + j\beta l)$$

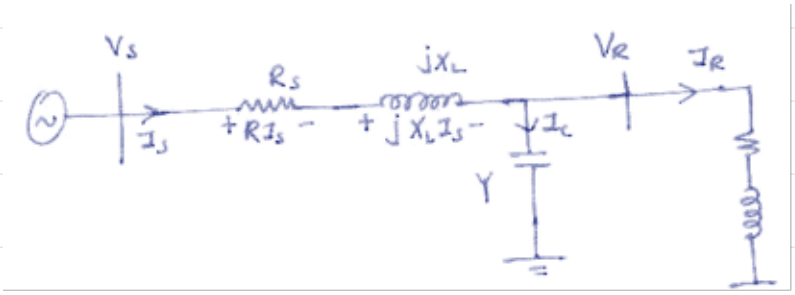
$$= Z_c \left[\sinh(\alpha l)\cos(\beta l) + j(\cosh(\alpha l)\sin(\beta l)) \right] \quad -(1)$$

For loss less line : $\alpha = 0$, $\beta = \sqrt{ZY}$, $\gamma = \alpha + j\beta = \sqrt{ZY}$, $Z_c = \sqrt{\frac{Z}{Y}} = \sqrt{\frac{L}{C}}$

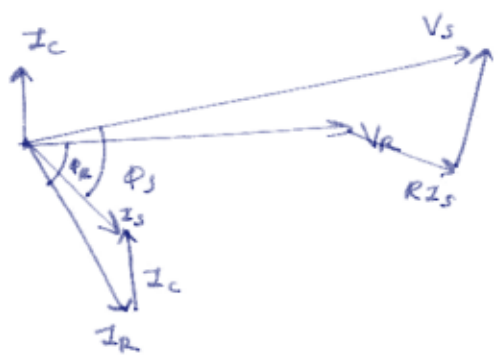
$$\boxed{Z_{eq} = \sqrt{\frac{Z}{Y}} \left[\sin\left(\sqrt{ZY} \times \frac{180}{\pi} \right) l \right]}$$

END CONDENSER N/W :

(a) with load :



Vs, Is - Rs, jXL (voltage drops R Is, j XL Is) - Ic through Y - VR, IR - lag load (cos phiR)



Obs:

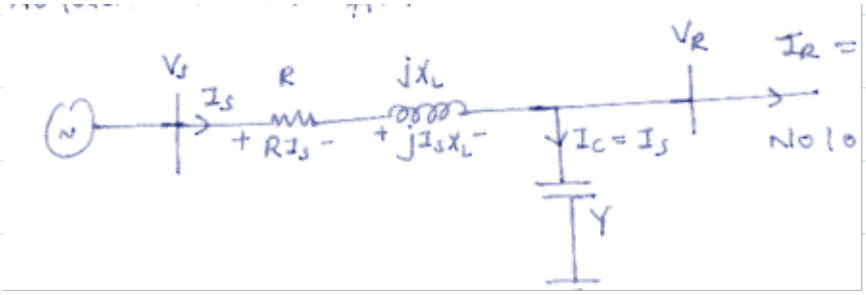
$|V_S| > |V_R|$ $V_{reg} > 0$

$|I_S| < |I_R|$ due to I_c

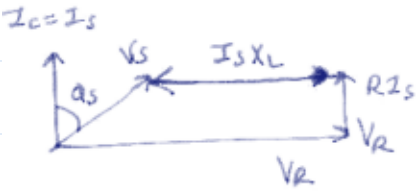
I_S lags V_S

Phasor diagram: Vs, VR, R Is, Ic, Is, IR; angles phiR and phiS. Obs: |Vs| > |VR|, Vreg > 0; |Is| < |IR| due to Ic; Is lags Vs

(b) No load : - Ferranti effect.



No load: IR = 0, Ic = Is



$V_R > V_S \rightarrow V_{reg} < 0$

$I_R = 0 \Rightarrow I_c = I_S$

I_S leads V_S

Phasor diagram: Ic = Is, Vs, Is XL, R Is, VR. VR > Vs, Vreg < 0; IR = 0 so Ic = Is; Is leads Vs

* Ferranti effect occurs in medium and long Tr. line under no load or off peak load or light load condition.
* Due to this effect VR > Vs occurs, and this overvtg at receiving end causes damage to the insulation of S/s equipment and Tr. line.
* To protect this S/s equipment, shunt reactors are practically used in the 400 kV, 765 kV S/s. These shunt reactors absorb excess lagging reactive power delivered by system capacitances so that VR = Vs (Vtg Regulation = 0) is maintained. These are practically kept in service during night or off peak load period.
* If the loading of the line is less than Surge Impedance loading (SIL), Ferranti effect occurs.

I_c, V_R at no load : At no load $I_R = 0$

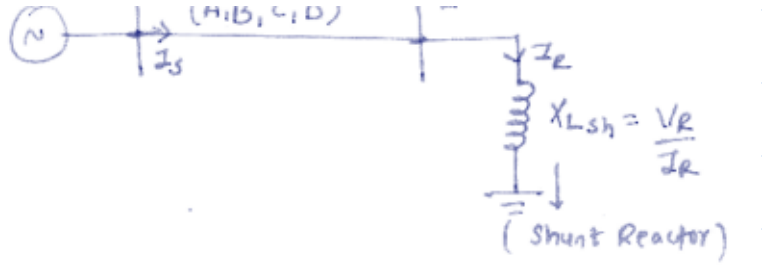
(1) $V_S = AV_R + BI_R \Rightarrow V_S = AV_R + B(0)$

$$\boxed{V_R = \frac{V_S}{A}}$$

(2) $I_S = CV_R + DI_R \Rightarrow I_S = I_c = CV_R + D(0)$

$$|I_c| = \left| C \frac{V_{RL}}{\sqrt{3}} \right|$$

Compensation of Ferranti effect : At no load $I_R = 0$, $V_R > V_S$. To make $V_R = V_S$ shunt reactors are connected at S/s.



Vs - line [A, B, C, D] - VR; shunt reactor XLsh = VR / IR

$$V_S = AV_R + BI_R \text{ with } X_{Lsh}: \; V_R = V_S = AV_R + BI_R \Rightarrow V_R(1-A) = BI_R$$

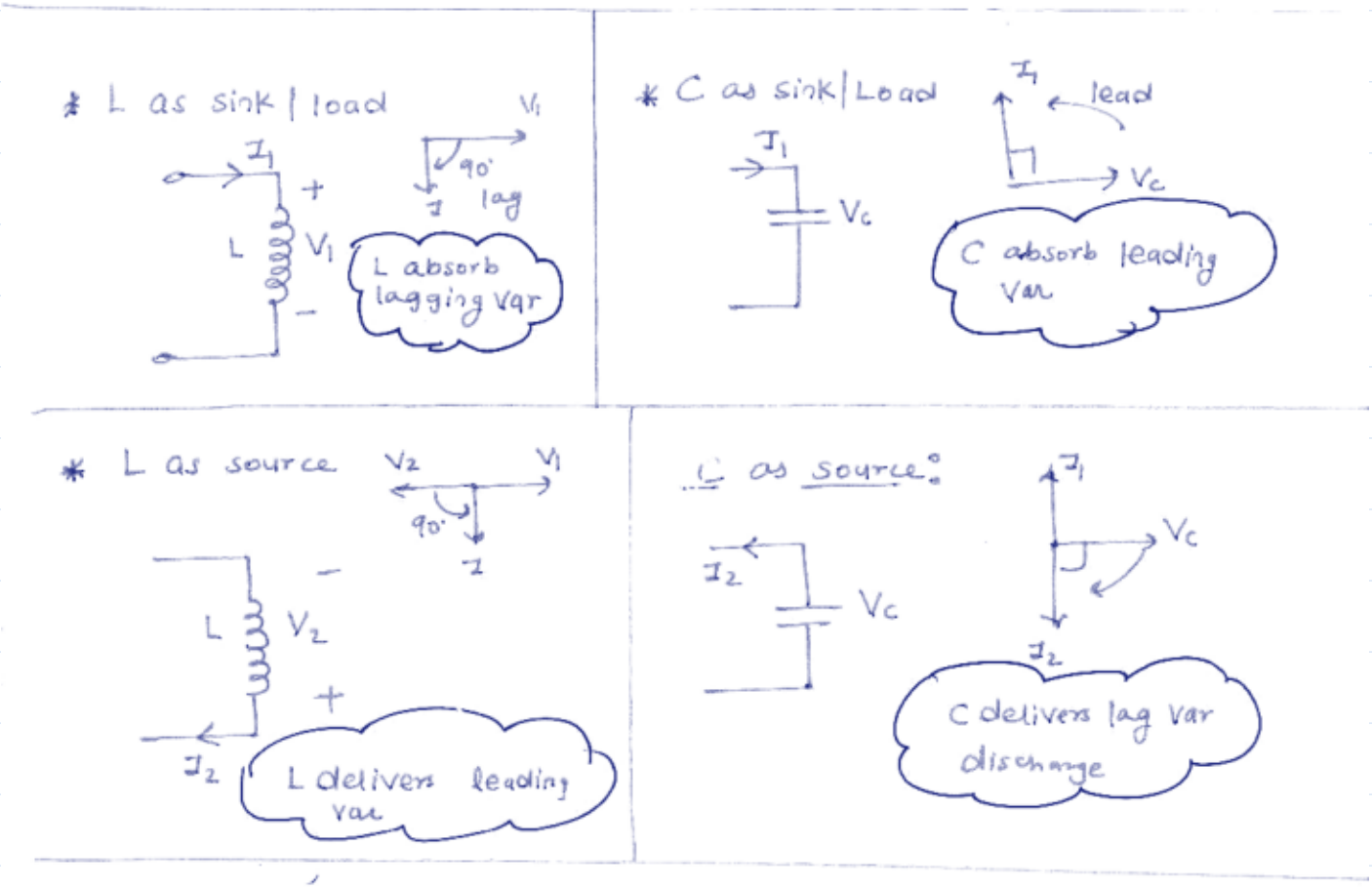
$$X_{Lsh} = \frac{V_R}{I_R} = \frac{B}{(1-A)} \quad \Omega/ph$$

* L as sink / load : I_1 lags V_1 by 90° => L absorbs lagging Var.

* C as sink / load : I_1 leads V_C => C absorbs leading Var.

* L as source : I_2 ... L delivers leading Var.

* C as source : C delivers lag Var (discharge).



L and C as sink (load) and as source: current direction, voltage polarity and phasor (90 degrees lag / lead)

Eg:

$$V_{RL} = \frac{V_{SL}}{A} = \frac{400}{0.8} = 500\,kV, \quad I_c = \frac{CV_{RL}}{\sqrt{3}} = \frac{0.5 \times 10^{-6} \times 500 \times 10^3}{\sqrt{3}} = 0.144\,A/ph$$

$$X_{Lsh} = \frac{B}{1-A} = \frac{100}{1-0.8} = 500\,\Omega$$

Eg: A 300 km lossless T/L operating at 50 Hz, V_s = 1 p.u. Calculate V_R at no load.

$$A = 1 - \frac{(\beta')^2}{2} = 1 - \frac{1}{2} \times \left(\frac{2\pi f l}{v}\right)^2 = 1 - \frac{1}{2} \times \left(\frac{2\pi \times 50 \times l}{v}\right)^2 = 0.9506$$

$$V_{RL} = \frac{V_{SL}}{A} = \frac{1}{0.9506} = 1.052\,p.u$$

Eg (4):

$$A = 1 - \frac{\beta'^2}{2} = 1 - \frac{0.5^2}{2} = 0.875$$

$$(2)\ V_{RL} = \frac{V_{SL}}{A} = \frac{400}{0.875}$$

$$I_c = \frac{CV_{RL}}{\sqrt{3}} = \frac{0.5 \times 10^{-6} \times 457 \times 10^3}{\sqrt{3}} = 0.131\ A$$

$$(4)\ X_{Lsh} = \frac{B}{1-A} = \frac{200}{1-0.875}$$

Eg: A two bus p.s shown in the figure supplies load of $1 + j0.5$ p.u

$$P_R = \frac{V_S V_R}{X_L}\sin\delta \ :\ 1 = \frac{V_1 \times 1}{0.1}\sin(0 - \delta_2)$$

$$Q_S = \frac{V_S V_R}{X_L}\cos\delta - \frac{V_R^2}{X_L} \ :\ 0.5 = \frac{V_1 \times 1}{0.1}\cos(0 - \delta_2) - \frac{1^2}{0.1} \quad -(2)$$

$$V_1 = 1.05, \quad \delta_2 = -5.44^\circ$$

Eg (7):

$$A = 1 - \frac{1}{2} \times \left(\frac{2\pi f l}{v}\right)^2 = \frac{V_S}{V_R} \Rightarrow l = 294.71\ km$$

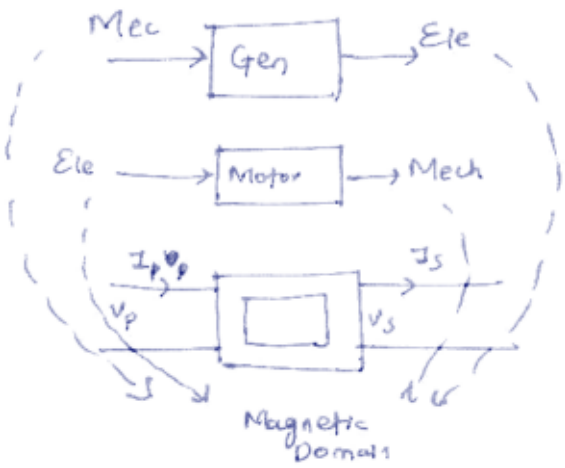
$$1 - \frac{1}{2} \times \left(\frac{2\pi \times 50 \times l}{3 \times 10^8}\right)^2 = \frac{400}{420} \Rightarrow l = 294.7\ km$$

* In 400 and 765 kV S/s, over-vtg relays are installed to protect the S/s equipments from O/V.

* Vtg and time grading are used for proper co-ordination between the lines and equipments.

* In normal vtg grading of the relay starts from 109% (436 kV for 400 kV lines) to 111% for 444 kV, and time gradation is 4 sec to 10 sec.

Importance of Reactive Power :



Gen: mechanical -> electrical; Motor: electrical -> mechanical; Transformer: Ip, Vp, Is, Vs through the magnetic domain